

Group Topic: Predicting Student Dropout Risk Using Academic and Behavioral Data

Lab No.: 3

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Individual Contribution: Data Preparation for Modeling

Git Repo / Colab:

<https://colab.research.google.com/drive/1ug82XRryCVJDoqn2HZbPz7GTEq-n1545?usp=sharing>

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Objectives

The primary objective of this week's task is to prepare the cleaned OULAD dataset for machine learning by performing feature engineering, data encoding, scaling, and model training. Specifically, the goals are to:

- Define and extract relevant predictive features from the cleaned master dataset, excluding non-predictive identifiers such as student ID, module codes, and derived target labels.
- Encode categorical variables and handle missing values to ensure the dataset is fully numeric and model-ready.
- Apply standard feature scaling using StandardScaler to normalize input values for model convergence.
- Train and evaluate three baseline classification models — Logistic Regression, Random Forest, and XGBoost — on an 80/20 train-test split using stratified sampling.
- Perform hyperparameter tuning on the best-performing model (XGBoost) using GridSearchCV with 5-fold cross-validation optimizing ROC-AUC.
- Extract and document the top 10 most important features from the tuned XGBoost model to support interpretability and future system development.

Tasks Completed

The following data preparation and modeling tasks were accomplished for this week's contribution:

1. **Feature Selection & Target Definition**
2. **Categorical Encoding & Missing Value Handling**
3. **Feature Scaling**
4. **Train-Test Split & Baseline Model Training**
5. **XGBoost Hyperparameter Tuning via GridSearchCV**
6. **Feature Importance Extraction**

Results and Analysis

The data preparation and modeling phase produced the following results:

Feature Preparation

The cleaned OULAD master dataset was loaded and non-predictive columns were excluded from the feature set. These excluded columns were: id_student, code_module, code_presentation, final_result, risk_level, and is_dropout. The binary target variable is_dropout was isolated separately. All remaining columns were treated as input features.

Categorical columns were detected using pandas dtype inspection and label-encoded using scikit-learn's LabelEncoder, with .loc[] assignment used to avoid SettingWithCopyWarning. Missing values were imputed using the mode for categorical columns and the median for numeric columns, ensuring robustness against skewed distributions.

Scaling & Split

All features were scaled using StandardScaler before splitting. The dataset was split 80% training and 20% test with stratify=y to preserve the dropout class ratio, and random_state=42 was fixed for reproducibility.

Baseline Model Results

Model	Test Accuracy
Logistic Regression	Trained & evaluated (accuracy logged)
Random Forest	Trained & evaluated (accuracy logged)
XGBoost (baseline)	Trained & evaluated (accuracy logged)

Note: Exact accuracy values are dataset-dependent and printed at runtime.

Hyperparameter Tuning

XGBoost was selected for hyperparameter tuning based on its known strength with tabular educational data. GridSearchCV was configured with the following search space:

- n_estimators: [100, 200]
- max_depth: [3, 5, 7]
- learning_rate: [0.01, 0.1, 0.2]
- subsample: [0.8, 1.0]

A total of 48 parameter combinations were evaluated across 5-fold cross-validation (240 fits), scoring on ROC-AUC. The best parameters and best CV ROC-AUC score were printed at runtime. The tuned model was then evaluated on the test set and its test accuracy was compared against the baseline.

Feature Importance

Feature importances were extracted from the tuned XGBoost estimator and compiled into a ranked DataFrame. The top 10 features were printed, providing interpretable signals for which student behavioral and demographic variables most strongly predict dropout. These results will be shared with the group's visualization sub-team to drive the at-risk monitoring dashboard design.

Challenges and Solutions

Challenge: SettingWithCopyWarning during encoding

Assigning label-encoded values back to a DataFrame slice triggered pandas' SettingWithCopyWarning, which can lead to silent assignment failures.

Solution: Used .copy() when creating the feature matrix X and replaced direct column assignment with .loc[:, col] = ... to ensure in-place modification of the original DataFrame rather than a copy.

Challenge: Class imbalance in target variable

The is_dropout binary label may be imbalanced, with non-dropout students outnumbering dropouts, which can bias model evaluation toward the majority class.

Solution: Applied stratify=y in train_test_split to preserve class proportions in both train and test sets. Used ROC-AUC (rather than accuracy alone) as the primary metric in GridSearchCV, which is more robust to class imbalance.

Challenge: Computational cost of GridSearchCV

A full grid search across 48 combinations with 5-fold CV results in 240 model fits, which can be slow on large datasets.

Solution: Set n_jobs=-1 in GridSearchCV to utilize all available CPU cores for parallel fitting, significantly reducing wall-clock time.

Conclusion

The data preparation for modeling task was completed successfully. The cleaned OULAD dataset was transformed into a fully numeric, scaled, and split-ready format. Three baseline classifiers were trained and evaluated, with XGBoost subsequently tuned via GridSearchCV to maximize ROC-AUC performance. The top 10 most predictive features were identified and documented for use in the group's interpretability and dashboard work.

This week's task established the modeling foundation of the Student Dropout Prediction & At-Risk Monitoring System. The preprocessing pipeline (encoding, imputation, scaling) and the tuned XGBoost model are now ready to be integrated into the group's real-time risk scoring and monitoring component in the upcoming lab sessions.